Elly Kirwa

Contact: kirwa@iastate.edu | https://github.com/EllyKp/ellykp.github.io | https://orcid.org/0009-0002-9413-7915

PROFESSIONAL SUMMARY

Veterinary epidemiologist and data scientist with 5+ years of experience applying statistical analysis and machine learning to complex health and production data. Experienced in building efficient data pipelines, developing interactive tools, and translating analytical results into actionable insights that support improved animal health, operational performance, and profitability.

EDUCATION

Ph.D. in Population Sciences in Animal Health, **Aug. 2023 – March. 2027 (Expected)**
Iowa State University (ISU), Ames, IA
Dissertation: "Applied Data Analytics for Breeding Herd Efficiency in Commercial Swine Production."

Graduate Certificate: Business Data Analytics **Jan 2026 - Dec 2026**

M.Sc. in Veterinary Preventative Medicine
ISU, Ames, IA **Aug. 2023 – Dec. 2025**

M.Sc. in Pharmacology & Toxicology
Nairobi University, Nairobi, Kenya **Sep. 2018 – Sep. 2022**

Doctor of Veterinary Medicine (DVM) **Sep . 2012 – Sep 2017**
Nairobi University, Nairobi, Kenya
With Distinction

SKILLS

- **Programming Languages**: R, Python, SQL, SAS.
- **Machine Learning**: Supervised/Unsupervised Learning, Deep Learning, Natural Language Processing.
- **Data Engineering**: Feature Engineering, Large-Scale Data Processing, Data Cleaning.
- **Epidemiological studies** and clinical trials
- **Cloud Services**: Microsoft Azure, DuckDb, AWS
- **Molecular techniques**: conventional polymerase chain reaction (PCR), DNA extraction, whole genome sequencing
- **Toxicology:** high-performance liquid chromatography (HPLC), gas chromatography (GC), spectroscopy
- **Personal Attributes**: Analytical, result-oriented, team leader, problem-solving, Oral and written, succinct, and logical communication
- **Clifton Strengths**: Strategic, Learner, Achiever, Responsibility

PROFESSIONAL EXPERIENCES

Data analytics. **Research Assistant, Iowa State University | Ames, IA** **Aug 2023 – Present**

- Precision Swine Management: Designing and implementing projects focused on developing algorithms to improve herd efficiency
- Data Analytics: Utilizing Python, R, and SAS for complex analysis of sow-level production data and sequenced data to develop benchmark performance dashboards and disease surveillance platforms.
- Field Trials: Designing and executing field trials assessing the impact of vaccination programs

***Veterinarian & Research Manager*, Stanley & Son Limited | Salama, Kenya** **June 2018 – May 2023**

- Data-Driven Production**:** Introduced production-guided research and data analysis to integrate evidence-based production efficiency, including feed quality testing and somatic cell count monitoring through a developed Farm Management System
- Antimicrobial Stewardship**:** Developed an antimicrobial surveillance platform, including a policy framework for antimicrobial use in animals and residues in milk, to ensure farm-level food safety.
- Clinical & Management**:** Managed all veterinary and animal husbandry requirements, including diagnosis, surgery, post-mortems, disease control programs, and staff supervision.

Africa Network for Animal Welfare – Nairobi, Kenya

Veterinary Intern **June 2017-May 2018**

- Attended all wildlife veterinary emergencies, rescues, treatments, field surgeries, and desnaring in various National Parks and Conservancies
- Facilitated humane termination of critically injured wildlife and appropriate disposal of carcasses
- Neutered about 330 animals in different spay/ neuter campaigns in partnership with local governments and VetTeks- USA
- Collected and processed blood samples on a variety of wildlife animals
- Evaluated animals' histories and determined which medications to administer based on their respective ailments
- Rabies control through vaccinations, spay/ neuter programs
- Initiated research-guided conservation model
- Reviewed Animal welfare laws in Kenya for consideration in a Parliamentary motion

One Health East and Central Africa /USAID -Kenya

OHCEA Workforce **June- July 2017**

- Executed demo site workforce on community needs assessment and data collection among the Maasai Community in Loitokitok area- Kenya
- Identified community One Health problems and priorities, ranked the problems using the matrix and linear rankings and provided simple solutions
- Educated the pastoral community on better livestock production methods and zoonotic diseases
- Developed a workforce guide model to be used in various One Health demo sites

Obulex Solutions Limited – **Westlands, Nairobi**

Product development officer- Part Time **May 2015- Dec 2022**

- Outsourced new clients and facilitated marketing research
- Assisted in the development and testing of various software, e.g. Revenue Information Management System (RMIS) for Isiolo and Mombasa Counties, Afya Poa Telemedicine App
- Graphics Design & Website development for different clients

AWARDS & FELLOWSHIPS

- Real Pork Scholar (2025), National Pork Board, USA
- Best oral PhD presentation at the ASAS Midwest conference, USA. – (2025)
- Mandela Washington Fellow (MWF**)** (2023)
- MSc Fellowship/Scholarship, University of Nairobi, 2017
- MSc Scholarship, Iowa State University &, 2023
- PhD Assistantship/Scholarship**,** Iowa State University &, 2023
- Best Overall Student, DVM Class of 2016/2017
- DVM Exchange Program Recipient, Free University of Berlin, Berlin, Germany, 2015
- Undergraduate Fellowship, University of Nairobi: One Health Research Project 2012-2017

PEER REVIEWED ARTICLES

i. **Kirwa, E**., Peterson, B., Grohmann, C. J., Silva, G. S., Ulguim, R. R., Mellagi, A. P., & Frizzo, M. (2025). 215 Assessment of the relationship between lactation feeding patterns, litter performance measures, and sow characteristics on sow farrowing efficiency metrics. *Journal of Animal Science*, *103*(Supplement_1), 50-52. https://doi.org/10.1093/jas/skaf102.055

ii. Musskopf, M.; Peterson, T.; Machado, I.; Tran Pham Tien, T.; **Kirwa, E**.; Carnevale de Almeida Moraes, D.; Cezar, G.; Mil-Homens, M.; Li, P.; De Conti, E.; Silva, A.P.P.; Holtkamp, D.J.; Linhares, D.C.L.; Silva, G.S. Assessment of the Relationship Between Bioexclusion Practices Applied in Wean-to-Harvest Sites and PRRS Outbreaks. Vet. Sci.2025, 12, 1000. https://doi.org/10.3390/vetsci12101000

iii. **Kirwa E**, Jon Tangen, Luz Belen Vargas, Mariah Musskopf, Tien Thinh Tran Pham, Ana Paula Silva, Steve Lawson Luis Gimenez, Giovani Trevisan[1], Daniel Linhares, Gustavo S. Silva (2025). Dynamics of PRRSV Seroconversion and immune-modulating properties of tilmicosin during naive gilt exposure to PRRS LVI

iv. **Kirwa, E**., Aboge G. O., Maitho, T. E., Mbindyo, C. M., Abuom, T. O., & Mainga, A. O. (2021). Antibiotic profile of Staphylococcus aureus and Coagulase negative *Staphylococci* species isolated from raw camel milk from Garissa County, Kenya. East African Journal of Science, Technology and Innovation, 2(4). https://doi.org/10.37425/eajsti.v2i4.364

v. Bahati D., Makau D., **Kirwa E**. & Mutura J. An assessment on community perspectives and views on natural resource management and its implications to biodiversity conservation 2019, Researchgate http://dx.doi.org/10.13140/RG.2.2.10344.34566

Abstracts, Articles & Presentations

i. **Kirwa EK,** Beau Peterson, Caleb Grohmann, Matt Frizzo, Rafael da Rosa Ulguim, Gustavo Silva. Predicting Farrowing Success to Inform Breeding Decisions in Commercial Swine Herds. Symposium on Artificial Intelligence in Veterinary Medicine, Ithaca, NY, USA, 2026. May 29th, 2026. Oral Presentation.

ii. **Kirwa EK,** Beau Peterson, Caleb Grohmann, Matt Frizzo, Rafael da Rosa Ulguim, Gustavo Silva. Leveraging Machine Learning to Predict Subsequent Farrowing Success: A decision support tool for sow management. Society of Animal Science, Midwest, Omaha, USA, 2026. March 9th, 2026. Oral Presentation.

iii. **Kirwa EK,** Cesar Moura, Ben Haberl, Swaminathan Jayaraman, Rafael da Rosa Ulgium, Gustavo Silva. Factors associated with gilt retention in breeding herds under one production system in the Midwestern United States, Society of Animal Science, Omaha, USA, March 10th, 2026. Oral Presentation.

iv. **Kirwa EK,** Cesar Moura, Ben Haberl, Swaminathan Jayaraman, Rafael da Rosa Ulgium, Gustavo Silva. Using Filed Data to Identify Drives of Gilt Retention and Lifetime Productivity, The Robust Pig Seminar, PIC, Feb 26th, 2026.

v. **Kirwa EK,** Beau Peterson, Caleb Grohmann, Matt Frizzo, Rafael da Rosa Ulguim, Ana Paula Mellagi, Gustavo Silva. Assessment of the relationship between lactation feeding patterns, litter performance measures, and sow characteristics on sow farrowing efficiency metrics, American Society of Animal Science, MidWest, Omaha, USA, 2025. March 10th, 2025. Oral Presentation.

vi. **Kirwa EK,** Cesar Moura, Swaminathan Jayaraman, Rafael da Rosa Ulgium, Gustavo Silva. Factors associated with gilt retention in breeding herds under one production system in the Midwestern United States, International Conference on Pig Livability Omaha, USA, 2025. Oral Presentation.

vii. **2025. Elly Kirwa**, Rafael Da Rosa Ulgium, Ana Paula Mellagi, Gustavo Silva, Beau Peterson, Caleb Grohmann, Matt Frizzo. Practical strategies to maximize sow efficiency. Available at: https://www.nationalhogfarmer.com/livestock-management/practical-strategies-to-maximize- sow-efficiency

viii. **2025. Elly Kirwa**, Rafael Da Rosa Ulgium, Ana Paula Mellagi, Gustavo Silva, Beau Peterson, Caleb Grohmann, Matt Frizzo- Linking lactation feeding patterns, litter performance, and sow characteristics to sow efficiency metrics. Benchmark. 2025 Spring Edition (printed version). Farms.com Media & Publishing & PigC HAMP, Inc. Page 6-7. https://www.pigchamp.com/flipbooks/benchmark-magazine/2025/USA/4/index.html

ix. Poeta Silva AP, Moraes DCA, Moura C, Johansen N, Musskopf MN, Nicolino R, **Kirwa EK**, Mil Homens MP, Tien TTP, Peng L, Arruda GA, Machado IF, Naseer R, Tong H, Will K, Rupasinghe K, Jayaraman S, Magalhaes E, Rademacher C, Zhang J, Trevisan G, Linhares DCL, Silva GS. Association of manure pumping practices with pig health outcomes in wean-to-market pig populations. 2024 ISU JAMES D. MCKEAN SWINE DISEASE CONFERENCE. July 23-24, 2024. Ames, Iowa State University of Science and Technology

x. Tien TTP, Zhang J, Mil-Homens MP, Dion K, Peng L, **Kirwa EK**, Musskopf MN, DeConti ER, Moraes DC, Cezar GA, Machado IF, Aljets E, Tong H, Guo B, Bedal LR, Hollis BL, Naseer R, Osemeke OH, Mowrer CL, Rademacher C, Silva GS, Trevisan G, Linhares DCL, Holtkamp D. Evaluation of readily available technologies for decontaminating PRRSV and PEDV from non-porous surface in truck cabins. 2024 ISU JAMES D. MCKEAN SWINE DISEASE CONFERENCE. July 23-24, 2024. Ames, Iowa

xi. **Kirwa EK**: One Health Concept - The Kenyan Perspective: Mandela Washington Fellowship Conference, Minneapolis, USA, 2023

xii. **Kirwa EK.,** Mbindyo, C. M., Abuom, T.: Phenotypic characterization of antimicrobial-resistant non-pathogenic Staphylococcus aureus, ESCMID/ASM Conference, Dublin, 2022. Poster Competition

xiii. **Kirwa EK:** How safe is our milk on the shelf? The breakdown in the farm-to-fork chain: Kenya One Health Conference, International Livestock Research Institute, Kenya, 2021

xiv. **Kirwa EK.,** Aboge G. O., Maitho, T. E., Mbindyo, C. M., Abuom, T. O., & Mainga, A. O. Genotypic characterization of antimicrobial resistant Staphylococcus aureus from raw camel milk: Kenyatta National Hospital Conference on Health, 2021

xv. **Kirwa EK.** The Kenyan Veterinary Student Charter, IVSA Conference - Pretoria, South Africa 2017

xvi. **Kirwa EK,** Bahati D., Makau D.,& Mutura J. An assessment of community perspectives and views on natural resource management and its implications to biodiversity conservation: Africa Network for Animal Welfare Conference -Nairobi, 2017

Manuscripts Under Preparation

Kirwa EK, Beau Peterson, Caleb Grohmann, Matt Frizzo, Rafael da Rosa Ulguim, Gustavo Silva. Early life traits as predictors of retention and lifetime performance in primiparous sows. Under preparation for the *PLOS journal*.

Grant Proposals.

Awarded:

Silva, Ximena Paz Portal, David Rosero, **Elly Kirwa**: Implementation of a real-time surveillance system for mortality in wean-to-finish lots -Pig Livability: $30,000

Silva, Rafael da Rosa Ulgium, **Elly Kirwa**: An Applied Framework for Early Identification of High-Risk Piglets in Commercial Swine Systems -Pig Livability: $30,000

Silva, Linhares, Holtkamp, Rosero, Trevisan, Rupasinghe, **Kirwa**: Development of a precision sow management system (PSMS) to improve longevity and productivity in breeding herds -ISU VPR: $50,000

On-Review

Silva, Linhares, Holtkamp, Rosero, Trevisan, Rupasinghe, **Kirwa**: Improving breeding herd efficiency by unravelling predictors of sow retention, productivity and profitability: USDA-CARE $300,000

Not Awarded:

Gustavo Silva, Giovani Trevisan, Daniel Linhares, Marcelo Almeida, Jianqiang Zhang,Luis Gimenez, Elly **Kirwa**: Evaluating PRRSV Infection Consistency After Live Virus Inoculation with Different Doses from Two Field Isolates - IPPA: $71,298

Gustavo Silva, Giovani Trevisan, Daniel Linhares, Marcelo Almeida, Jianqiang Zhang,Luis Gimenez, Elly **Kirwa:** Evaluating PRRSV Infection Consistency After Live Virus Inoculation with Different Doses from Two Field Isolates- NPB: $82,830

Gustavo Silva, Giovani Trevisan, Daniel Linhares, Marcelo Almeida, Jianqiang Zhang,Luis Gimenez, Elly **Kirwa:** Assessing genetic diversity and evolutionary dynamics of PRRSV following post-outbreak PRRSV management strategies of MLV, LVI or combination of both in commercial breeding herds. NPB: $96,100

LEADERSHIP & COMMUNITY ENGAGEMENT

Moderator- Weekly Swine Seminar

- *VetMed, Iowa State University* **Dec. 2025 – Present**

International Veterinary Students Association (IVSA)

- *Kenyan Chapter - IVSA Officer* **Jan 2016 - Sep 2017**

One Health Club, University of Nairobi

- *Assistant Secretary* **March 2014 - May 2018**

Kenyan Animal Welfare Advocacy Network, University of Nairobi

- *Member* **Dec 2016 - May 2017**

Other Affiliations (Member):

- Association of American Swine Veterinarians (AASV)
- American Society of Animal Science
- Toxicology Society of Kenya
- Kenya Small and Companion Animal Veterinary Association, Kenya
- Kenya Veterinary Association
- Kenya Society for the Protection and Care of Animals